Reasons for Assigned Scores

1) *An Architecture for Integrating Large Language Models with Digital Twins and Automation Systems*

Q1: Problem Definition (Score: 2)

Where:

- **Page 1: Section I. INTRODUCTION**
- Paragraph beginning with *"Traditional industrial automation systems…"* and the bullet-point list of core questions.

Q2: Problem Context (Score: 2)

Where:

- **Page 1: Introduction**
- **Page 2–3: Conceptual Architecture**
  Describes industrial automation pyramid, digital twin definition, system structure.

Q3: Research Design (Score: 2)

Where:

- **Pages 2–5: Sections II & III & IV**
  - Figures 1–4 (system architecture, digital twin integration)
  - Three design paradigms (snapshot, event-log, planning)

Q4: Contributions (Score: 2)

Where:

- **Page 1: Paragraph beginning "This paper makes the following contributions:"**

Q5: Insights (Score: 2)

Where:

- **Pages 6–7: Section VI. DISCUSSION**
  Includes “Application goals”, “Return on Intelligence”, “Integration Levels”, “Test-driven development”.

Q6: Limitations (Score: 1)

Where:

- **Pages 6–7: DISCUSSION**
  Contains limitations *implicitly* (e.g., negative ROI, need for further work) but **no explicit limitations section**.

2) Improving Adaptability in OptimizationBased- Decision Support Systems Through Large Language Models

Q1: Problem Definition (Score: 2)

Where:

- **Page 1: Abstract + Introduction**
  Clear articulation of DSS rigidity and need for flexibility.

Q2: Problem Context (Score: 2)

Where:

- **Pages 1–3: Introduction + Literature Review**
  Explains DSS limitations, case studies of failed DSS, cites optimization literature.

Q3: Research Design (Score: 2)

Where:

- **Pages 4–6: Section III. Framework**
- **Algorithm 1 (Page 6)**
- **Pages 6–13: Experiments C1–C6**

Q4: Contributions (Score: 2)

Where:

- **Page 1–2: Introduction**
  Explains NL instructions, dynamic reconfiguration, preference based decisions, explainability-.

Q5: Insights (Score: 2)

Where:

- **Pages 7–13: Section IV. Experiments**
  Prompt clarity degradation, robustness, scaling behavior, interpretability.

Q6: Limitations (Score: 2)

Where:

- **Pages 13–14: Section IV-C and Conclusion**
  Explicit discussion on non-scalable MCDM, need for iterative nondominated comparison, domain generalization.

3) *An Adaptive MultiAgent -LLMBased- Clinical Decision Support System Integrating Biomedical RAG and Web Intelligence*

Q1: Problem Definition (Score: 2)

Where:

- **Pages 1–2: Introduction**
  Clear clinical challenges: noisy data, evolving evidence, hallucinations, need for grounded reasoning.

Q2: Problem Context (Score: 2)

Where:

- **Pages 1–3: Introduction**
  Survey of CDSS systems, RAG, ontology systems, LLM limits, clinical reasoning requirements.

Q3: Research Design (Score: 2)

Where:

- **Pages 3–6: Methodology**
  Five agents, roles, architecture diagrams (Figures 2–5), BioBERT-Qdrant pipeline, feedback loop design.

Q4: Contributions (Score: 2)

Where:

- **Page 3: Contribution statements**
  Three key contributions listed explicitly.

Q5: Insights (Score: 2)

Where:

- **Pages 10–12: Experiments, ablation study, performance analysis**
  Trend analysis, diagnostic improvement, bottlenecks, agent-level contribution.

Q6: Limitations (Score: 2)

Where:

- **Pages 12–13: Section IV. LIMITATIONS AND DISCUSSION**
  Dedicated limitations discussing dataset representativeness, computational cost, hallucination risk, need for multicenter validation.